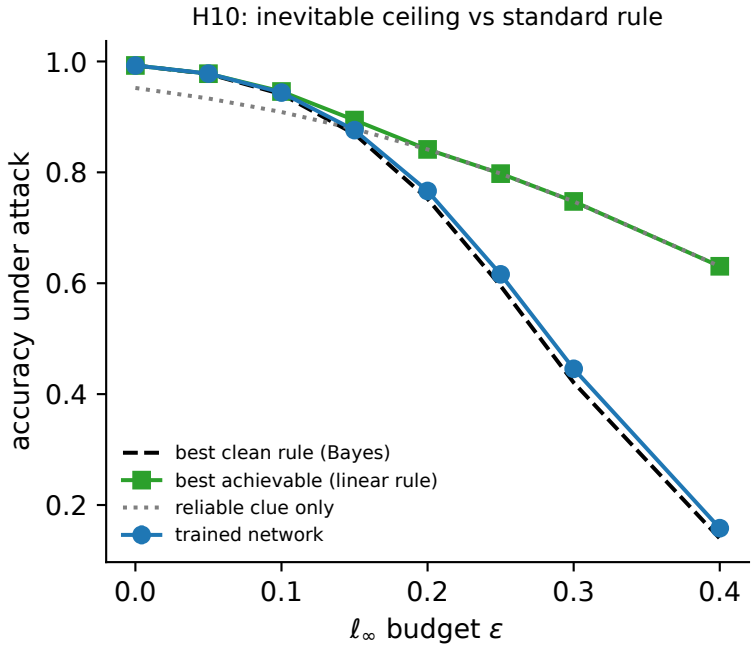
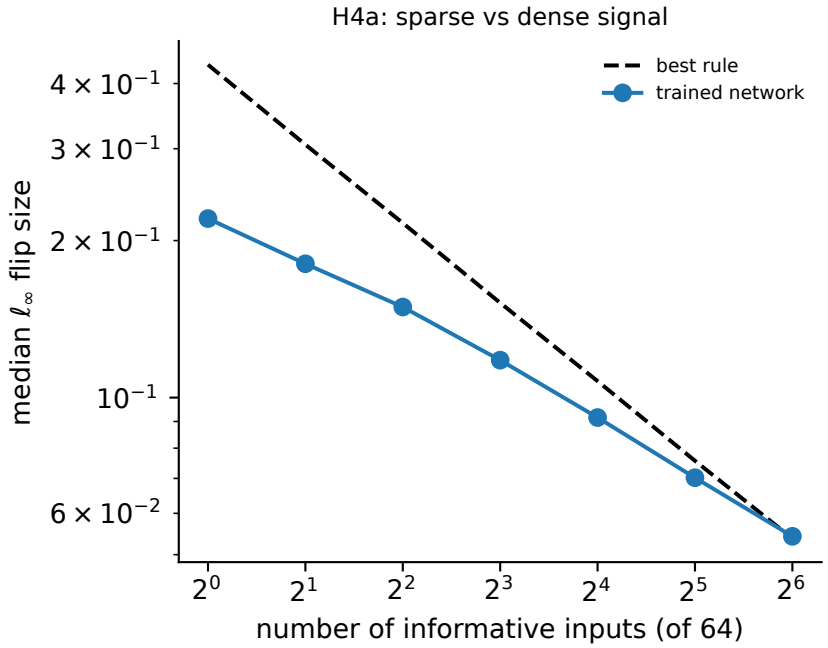
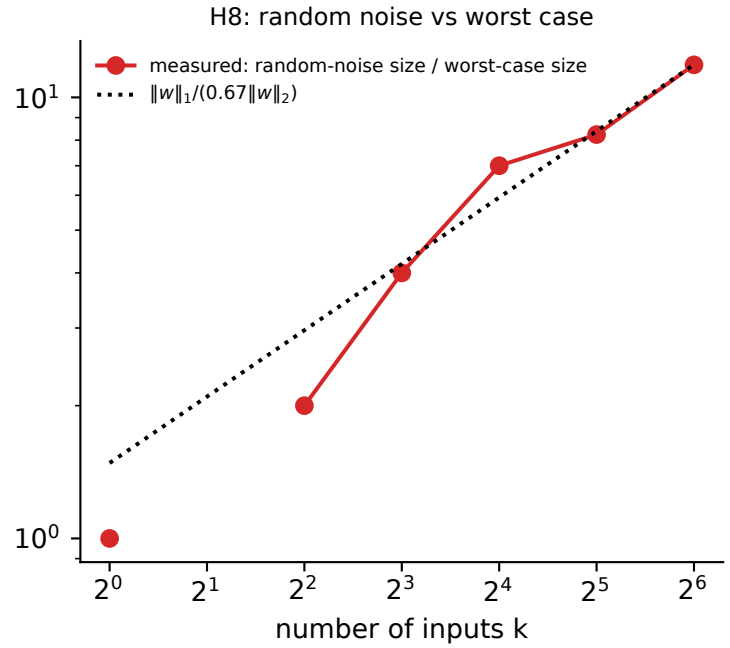
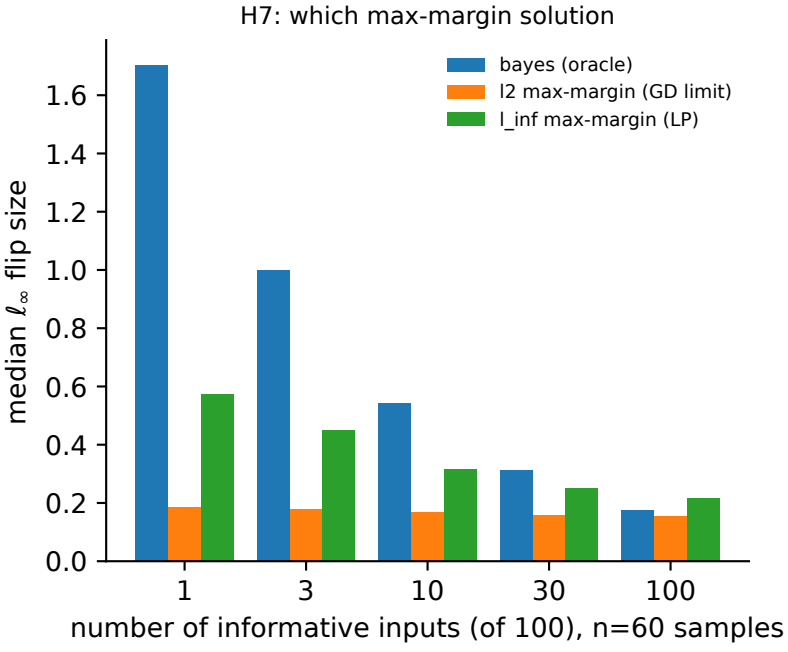
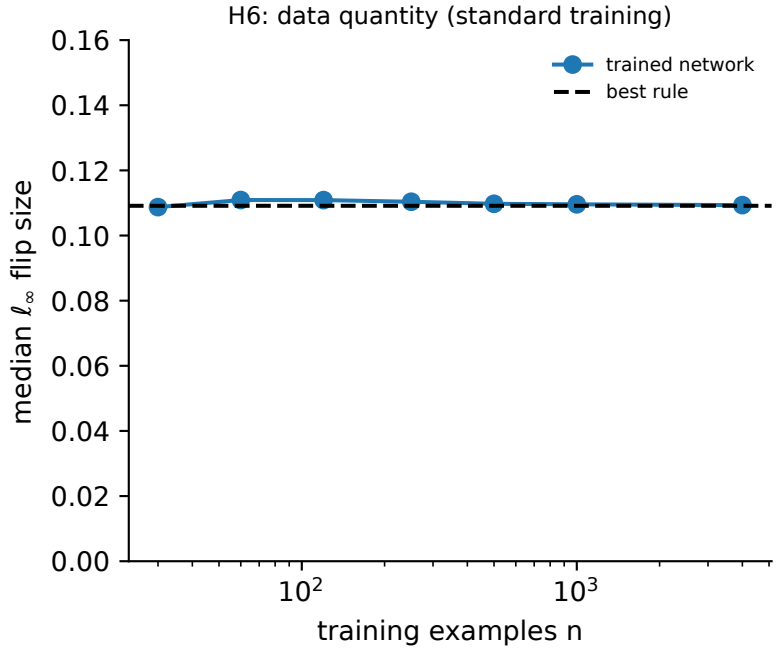
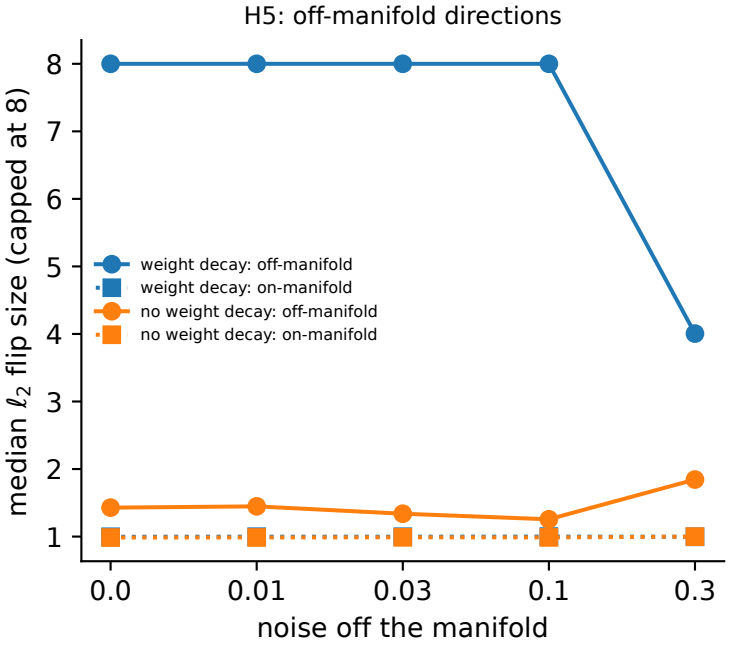

H10: inevitable ceiling vs standard rule
accuracy under attack
1.0
0.8
0.6
0.4
0.2
best clean rule (Bayes)
best achievable (linear rule)
reliable clue only
trained network
0.0
0.1
0.2
0.3
0.4
ℓ∞ budget ε


H4a: sparse vs dense signal
median ℓ∞ flip size
4 × 10⁻¹
3 × 10⁻¹
2 × 10⁻¹
10⁻¹
6 × 10⁻²
best rule
trained network
2⁰
2¹
2²
2³
2⁴
2⁵
2⁶
number of informative inputs (of 64)


H8: random noise vs worst case
10¹
10⁰
measured: random-noise size / worst-case size
‖w‖₁/(0.67‖w‖₂)
2⁰
2¹
2²
2³
2⁴
2⁵
2⁶
number of inputs k


H7: which max-margin solution
median ℓ∞ flip size
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
bayes (oracle)
l2 max-margin (GD limit)
l_inf max-margin (LP)
1
3
10
30
100
number of informative inputs (of 100), n=60 samples


H6: data quantity (standard training)
median ℓ∞ flip size
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
trained network
best rule
10²
10³
training examples n


H5: off-manifold directions
median ℓ2 flip size (capped at 8)
8
7
6
5
4
3
2
1
weight decay: off-manifold
weight decay: on-manifold
no weight decay: off-manifold
no weight decay: on-manifold
0.0
0.01
0.03
0.1
0.3
noise off the manifold